Lesson 3.7 Teaching Support Guide

Teach-tomorrow route

Print: current Guided Notes and Practice A for the class; Exit Tickets; optional Language Support and supported homework. Keep this guide and answer keys teacher-facing. Use the full-size model pages in the updated notes.

Phase	Planned minutes	Teaching move
Retrieve	5	Warm-Up items 1–4; note the missing prerequisite before launch.
Launch	7	Name the session target and invite a prediction from the conceptual task.
Model	10	Link a visual model to the worked example; rehearse the mathematical explanation.
Practice	18	Core: A #1, #2, #4, #6 and the error task. Choose remaining items from evidence.
Check	5	Exit Ticket; use the diagnostic page before assigning reteaching.

This is a planned 45-minute route, not a measured classroom duration. For a shorter lesson, keep the model and exit; reduce repeated practice. Form B is parallel practice or reassessment only when its items are unseen.

S1: Customary vs. Metric Conversions · I can convert a measurement between the customary and metric systems using a given rate.

S2: Multi-Step System Conversions and Dosage · I can solve a multi-step problem that requires converting units before using a rate.

Spaced retrieval: revisit one item after 2 lessons and after approximately 1 week. Mix it with the next unit’s prerequisite check; do not announce the operation before students choose a method.

Session 1 · Customary vs. Metric Conversions

I can convert a measurement between the customary and metric systems using a given rate. · 6.RP.A.3d

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 44 lb b. 10 mi	Build the given equivalent pair and scale it to the target while keeping ≈ visible.
2	6 × 3.28 = 19.68 ft	Build the given equivalent pair and scale it to the target while keeping ≈ visible.
3	A — multiply by 3.28	Ask the learner to show why A — multiply by 3.28 is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Uses an exact-equality claim for an approximate cross-system factor or multiplies in the wrong direction..

Response: Build the given equivalent pair and scale it to the target while keeping ≈ visible.

Also check: Dividing when the new unit is smaller (50 mi ÷ 1.6 = 31.25 km).

Also check: Comparing amounts in different units, such as 70 m and 160 ft, without converting.

If calculation alone is wrong, use Session 1 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 1 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

Using the same factor, convert 19.8 pounds to kilograms and check.

Teacher solution: Approximately 9 kg; 9 × 2.2 = 19.8 lb using the supplied approximation.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: 15.4 ÷ 2.2 = 7, so 15.4 lb ≈ 7 kg. A kilogram corresponds to more than one pound, so the numerical measure decreases. Check 7 × 2.2 = 15.4 using the approximate factor.

Misconception response anchor: Uses an exact-equality claim for an approximate cross-system factor or multiplies in the wrong direction.. Use build the given equivalent pair and scale it to the target while keeping ≈ visible.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: The mass is approximately ___ because ___; I use ≈ since ___.

Session 2 · Multi-Step System Conversions and Dosage

I can solve a multi-step problem that requires converting units before using a rate. · 6.RP.A.3d

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 25 kg b. 50 mg	Label the substance as well as the unit at each step, and match each rate's input to the previous output.
2	180 ÷ 12 = 15 L; 15 × $2 = $30	Label the substance as well as the unit at each step, and match each rate's input to the previous output.
3	A — before you use the rate	Ask the learner to show why A — before you use the rate is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Uses the rate before converting its reference quantity or confuses grams of pigment with grams of clay..

Response: Label the substance as well as the unit at each step, and match each rate's input to the previous output.

Also check: Using pounds with a per-kilogram rate (66 lb × 4 mg instead of 30 kg × 4 mg).

Also check: Multiplying by 2.2 to change pounds to kilograms.

If calculation alone is wrong, use Session 2 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 2 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

There are 17.6 lb of clay, 2 g pigment per kg, and 4 g per scoop. Using 1 kg ≈ 2.2 lb, find scoops.

Teacher solution: 17.6 ÷ 2.2 = 8 kg; 8 × 2 = 16 g; 16 ÷ 4 = 4 scoops, approximately.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: 11 lb ≈ 5 kg; 5 × 3 = 15 g pigment; 15 ÷ 5 = 3 scoops. The unit path is pounds → kilograms → grams pigment → scoops. The result is approximate because the conversion factor is approximate.

Misconception response anchor: Uses the rate before converting its reference quantity or confuses grams of pigment with grams of clay.. Use label the substance as well as the unit at each step, and match each rate's input to the previous output.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: I first convert ___ to ___ so that the rate ___ can be applied. Then ___.